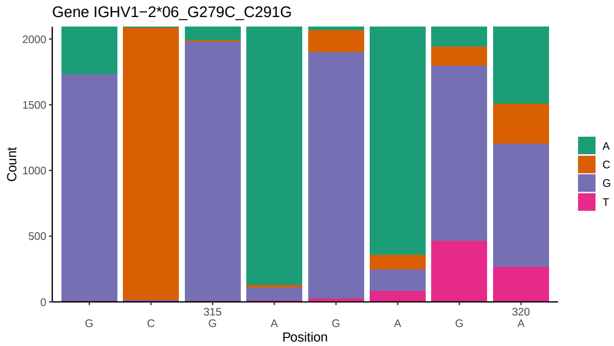
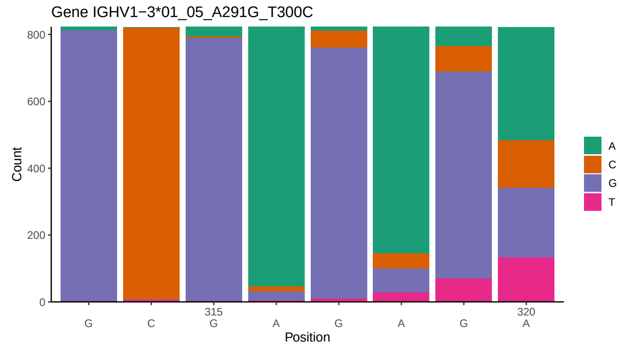
OGRDBstats Report

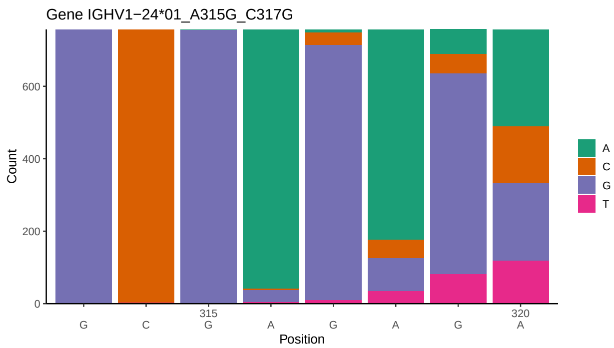
Contents

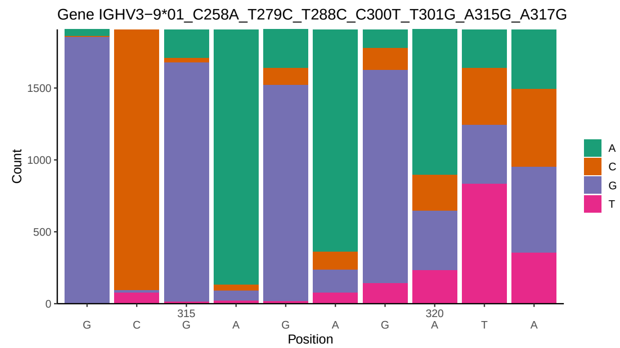
1 Novel sequence analysis

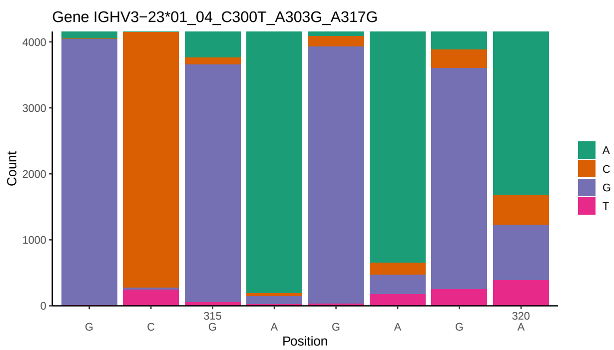
1.1 End-nucleotide composition

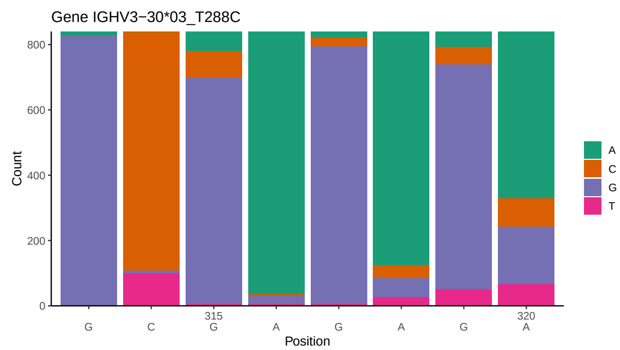


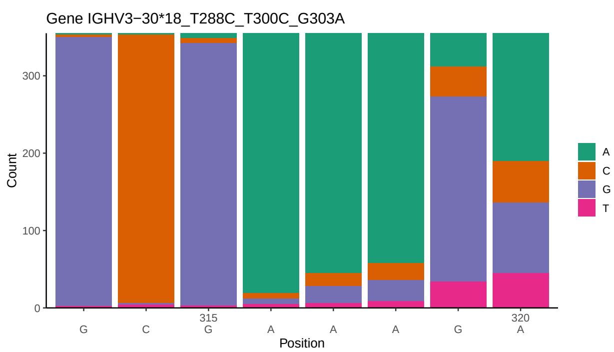


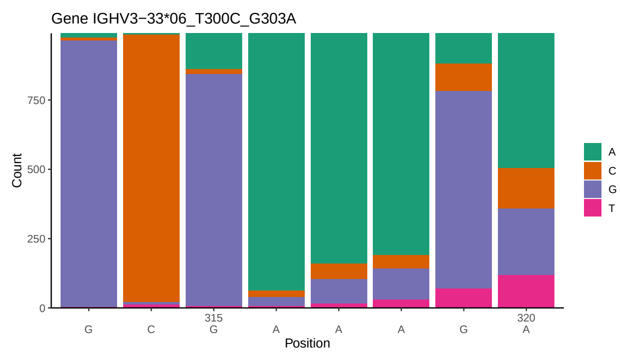


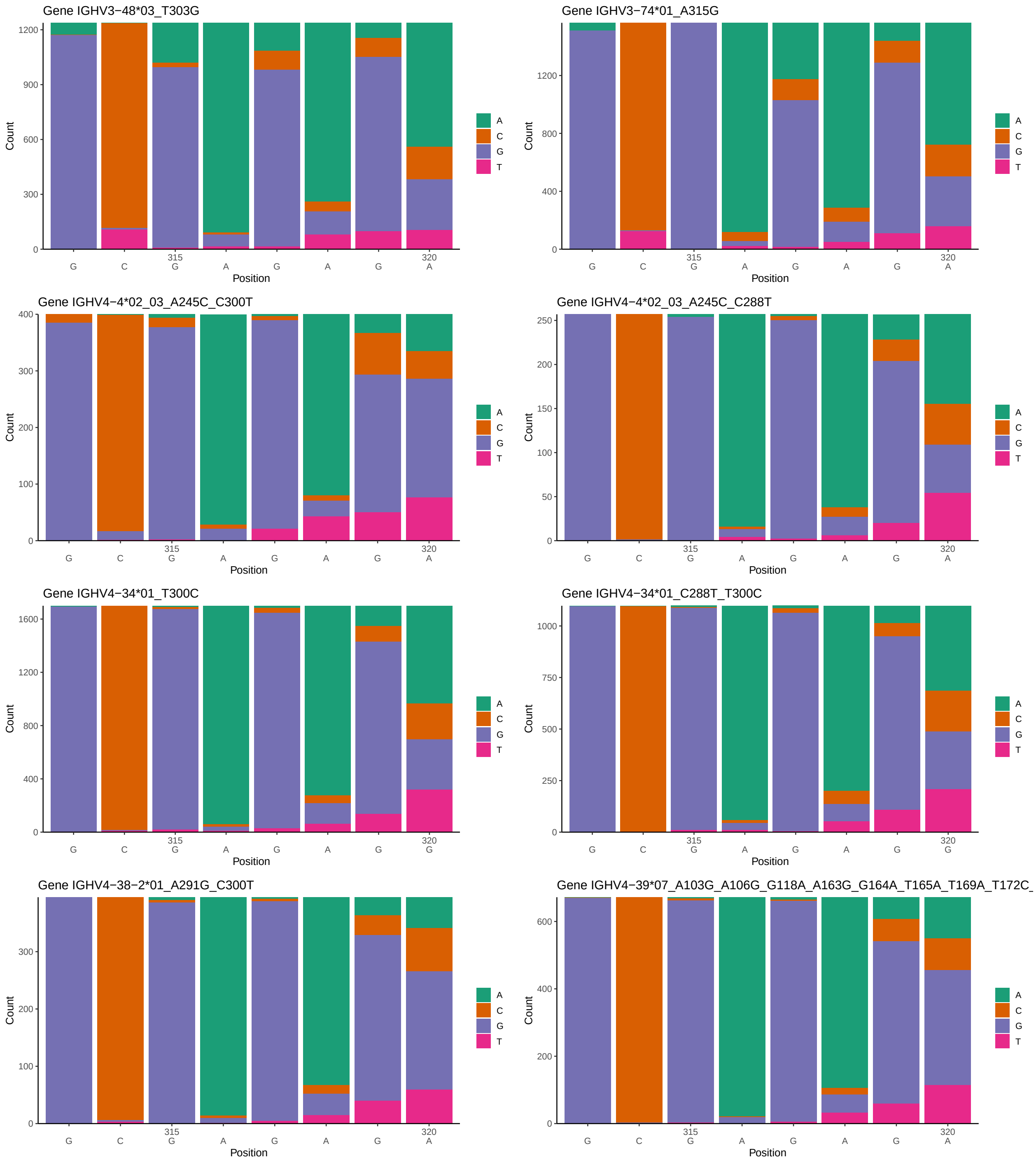




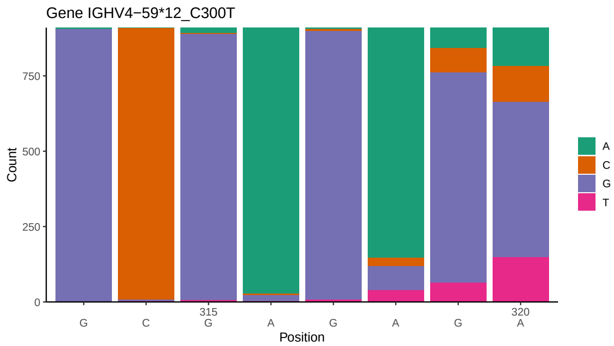


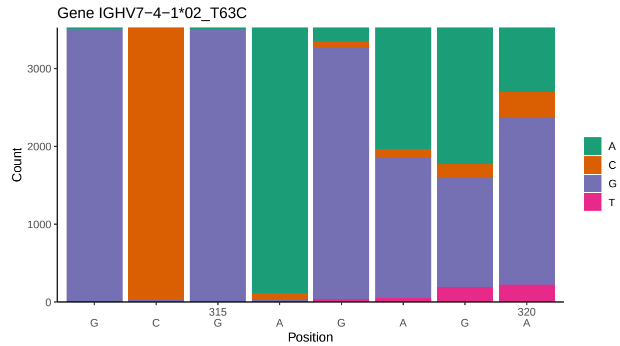




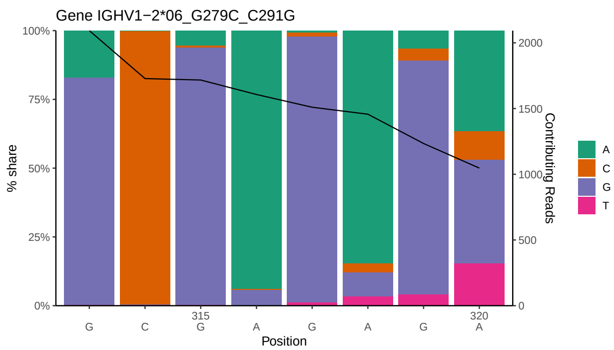


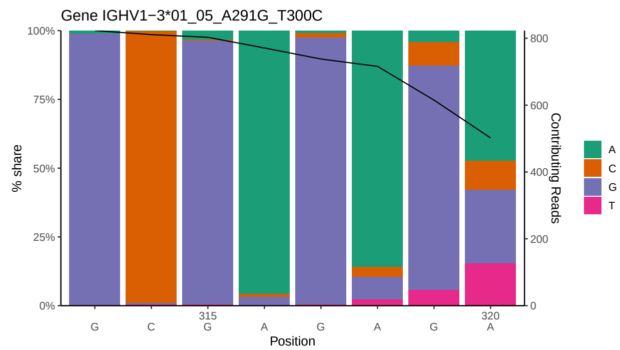
Gene IGHV3−48*03_T303G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−4*02_03_A245C_C288T
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−38−2*01_A291G_C300T
Gene IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C_
Count
Position
A
C
G
T

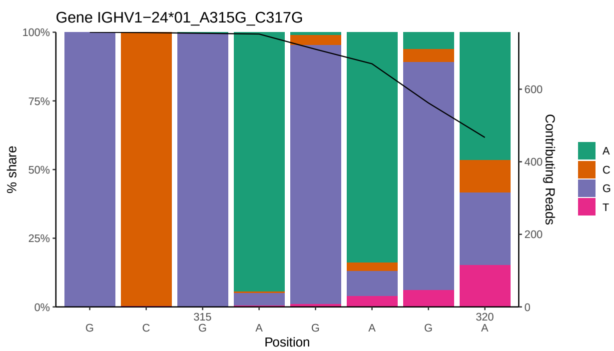


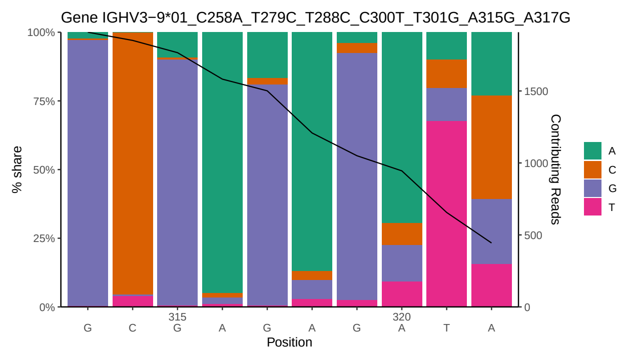


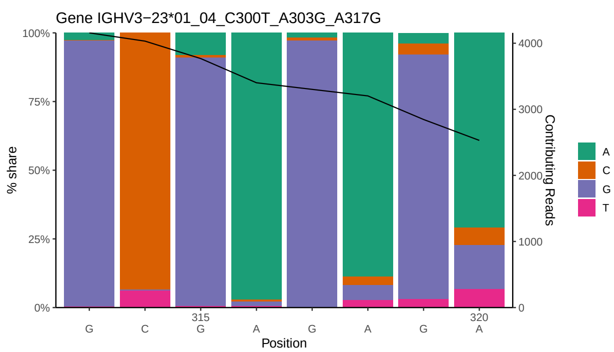
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

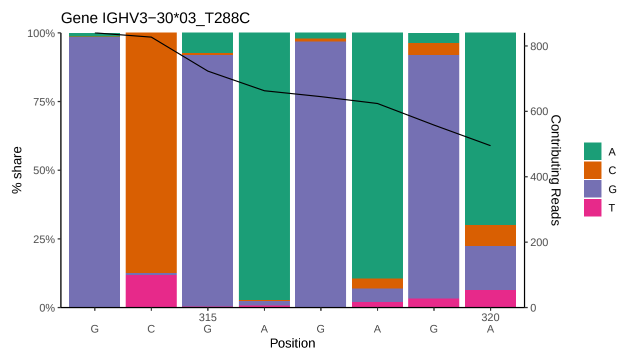


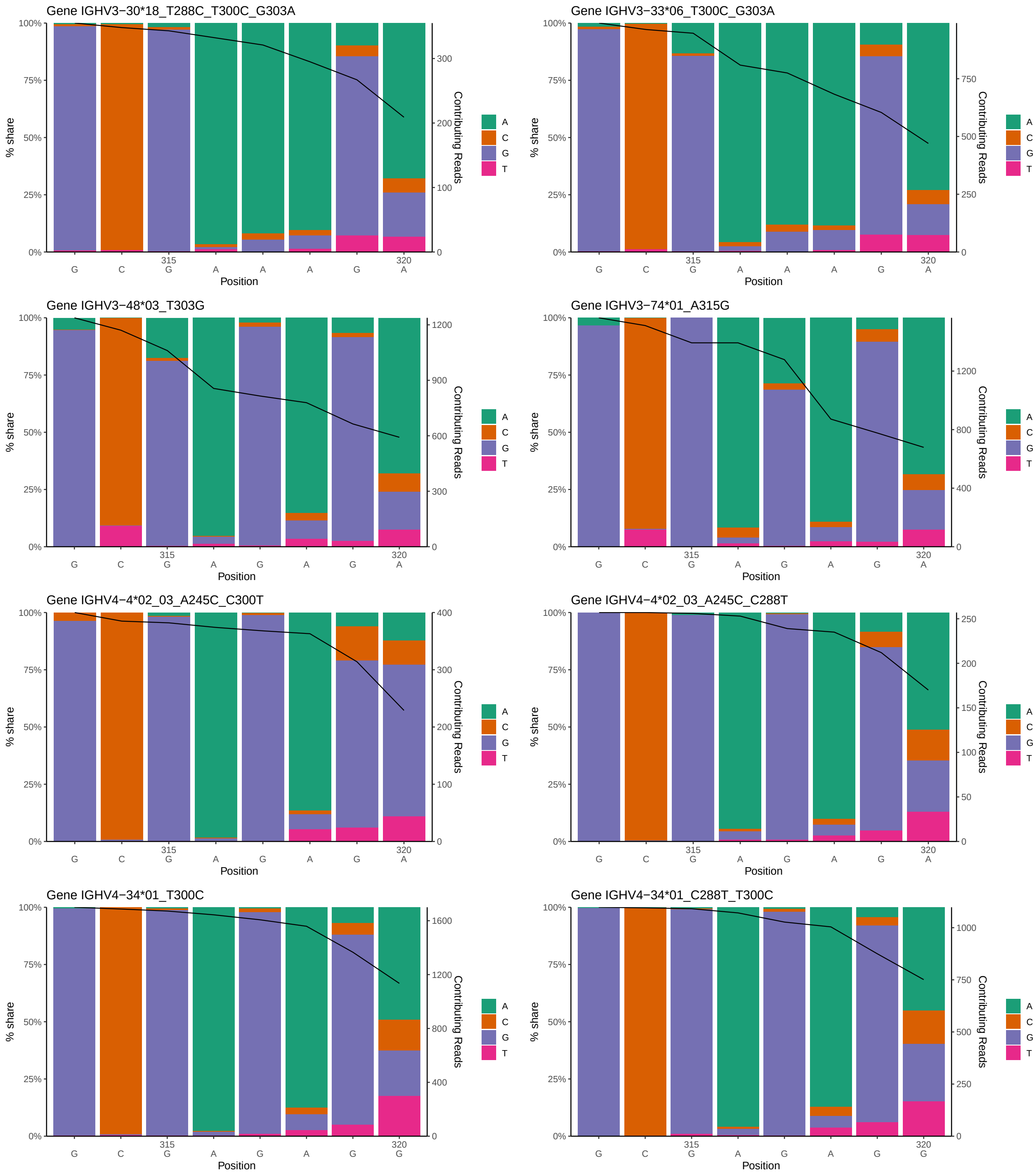


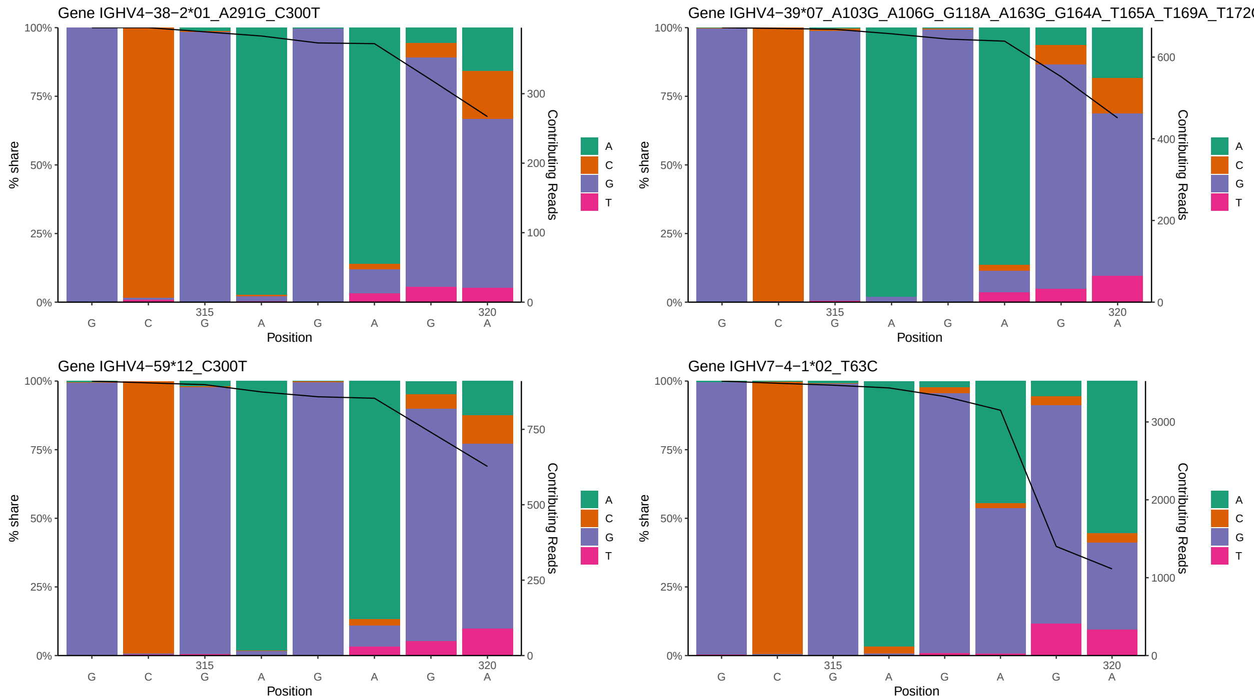




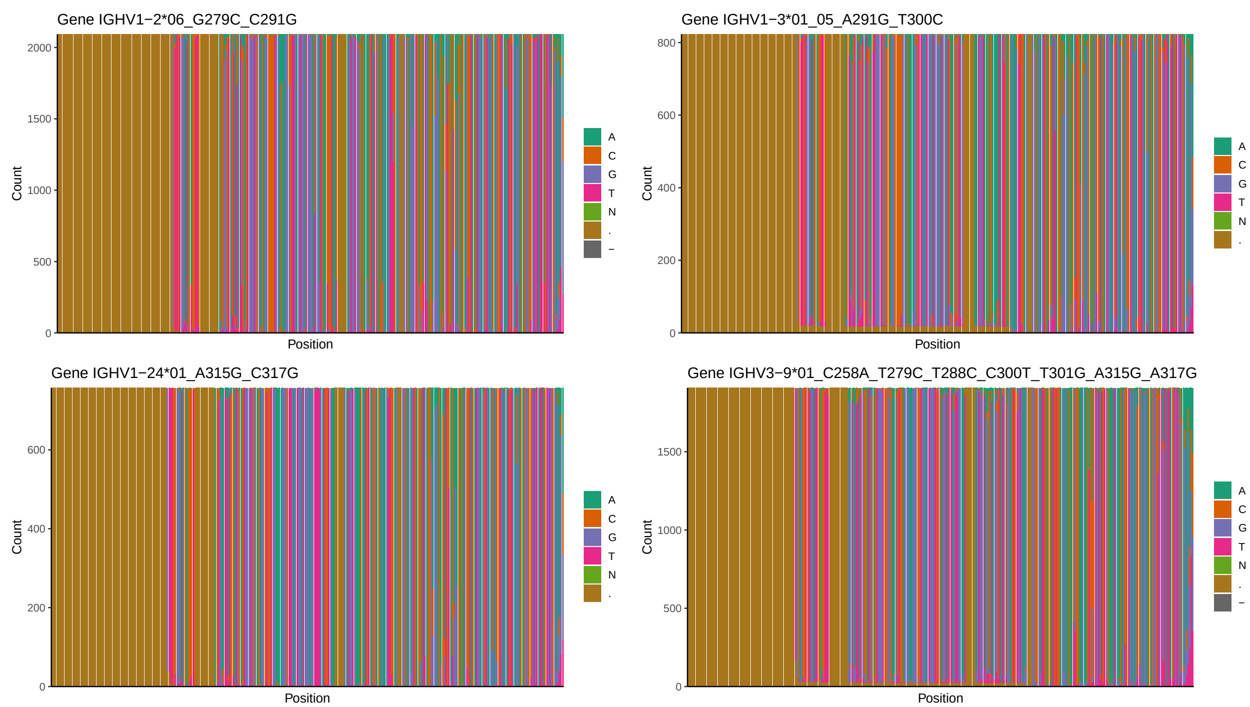


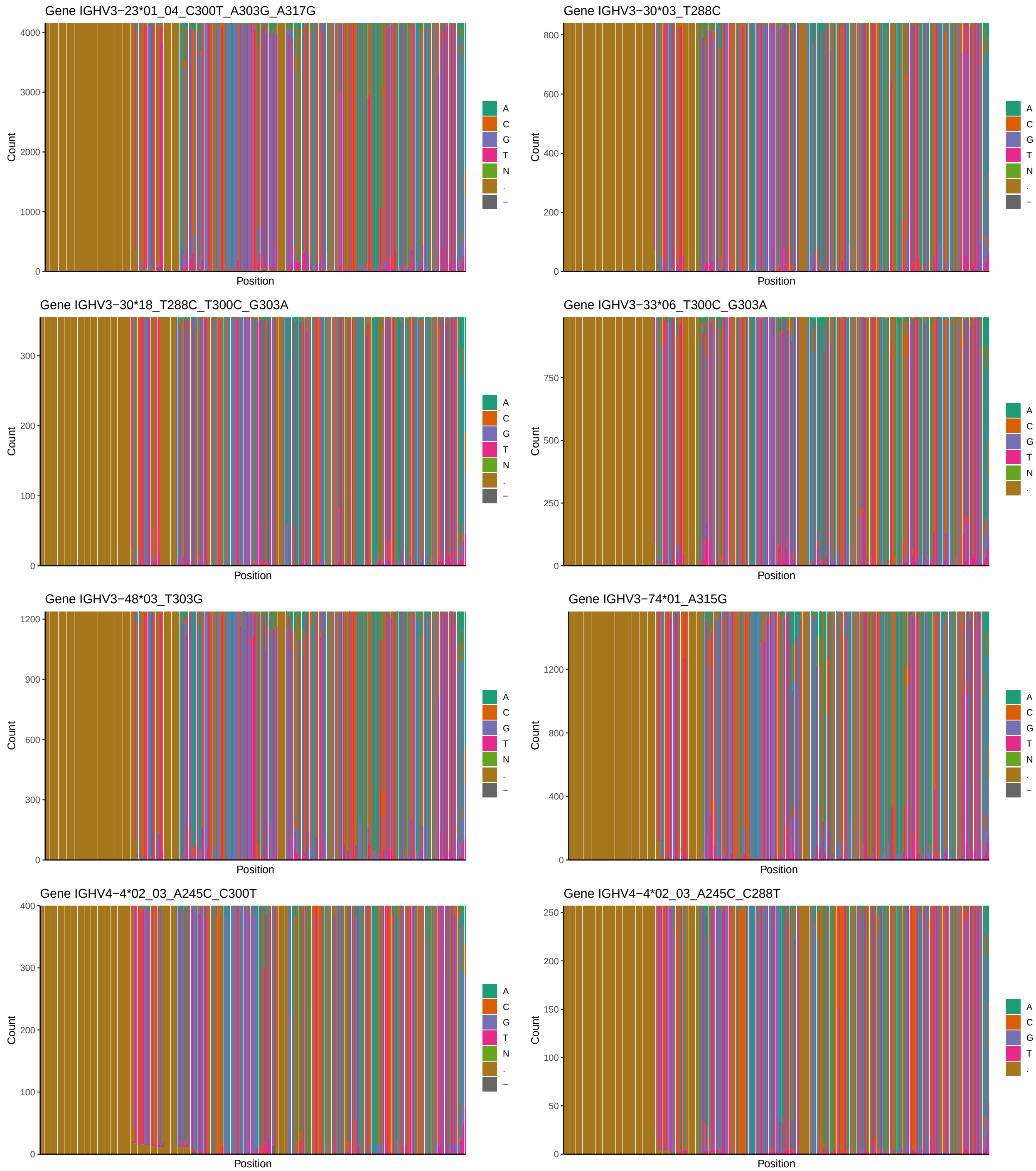


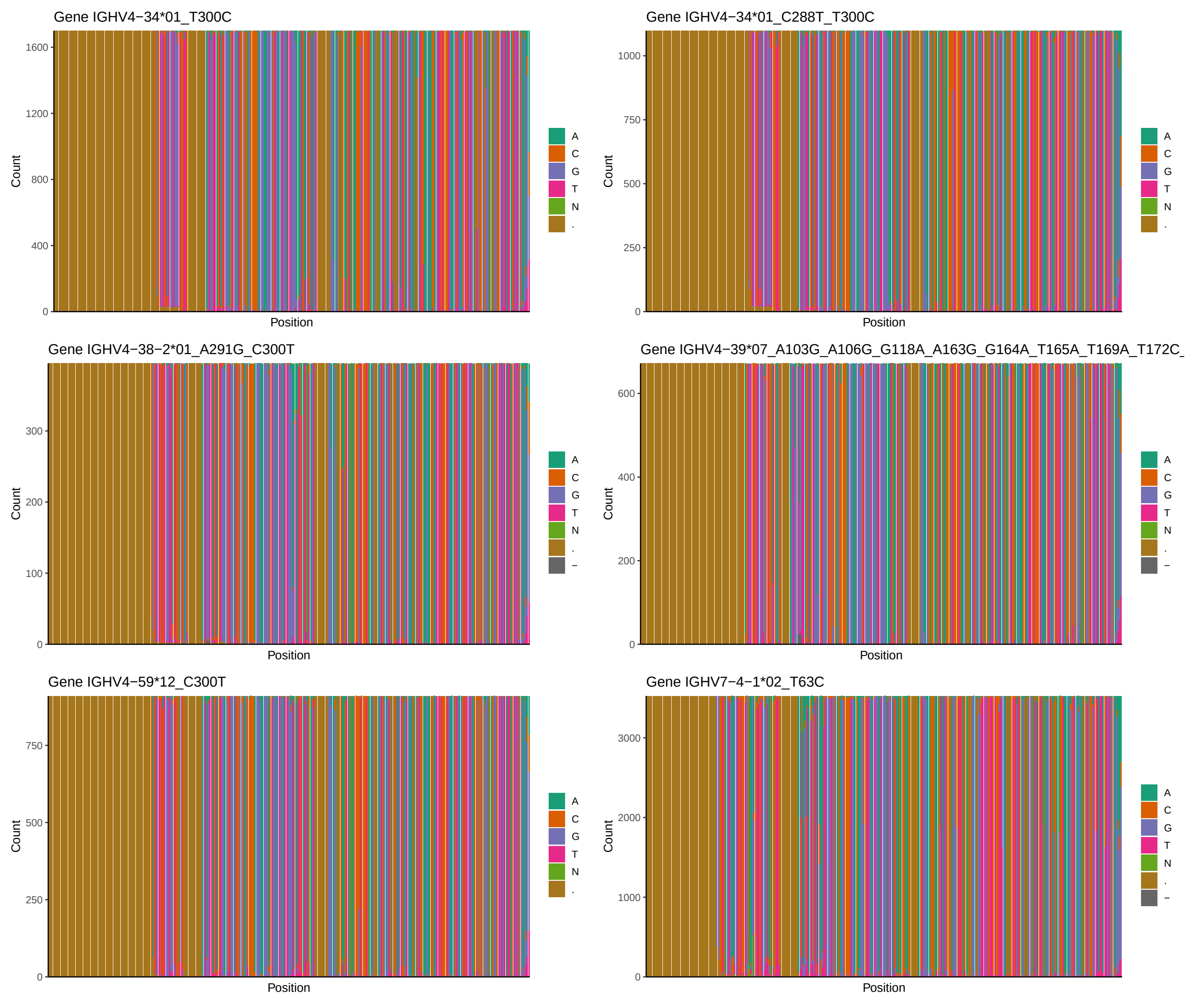

Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−4*02_03_A245C_C288T
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
% share
Contributing Reads
Position
A
C
G
T



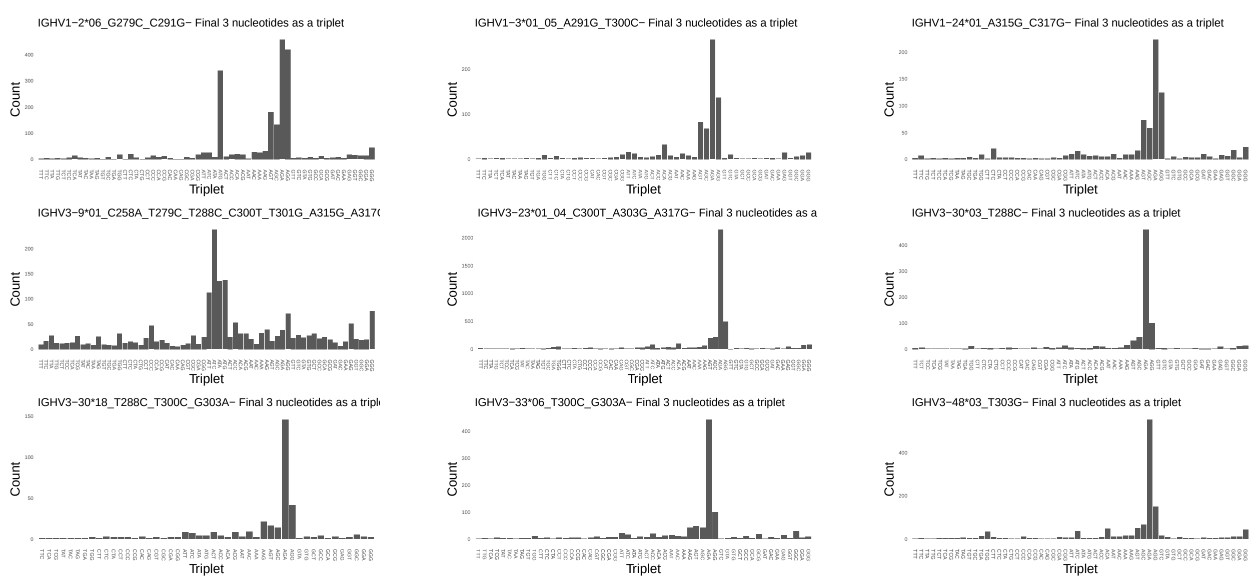
1.3 Whole-sequence composition of each assigned read

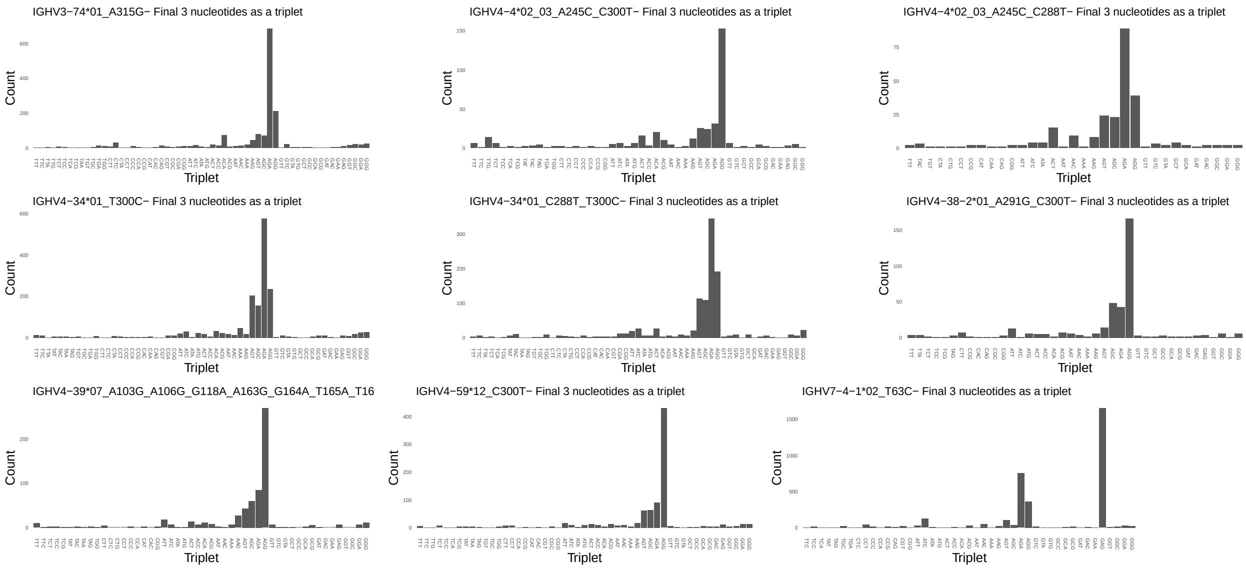



Gene IGHV3−23*01_04_C300T_A303G_A317G
Gene IGHV3−30*03_T288C
Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−4*02_03_A245C_C288T
Count
Position
A
C
G
T
N
.
−

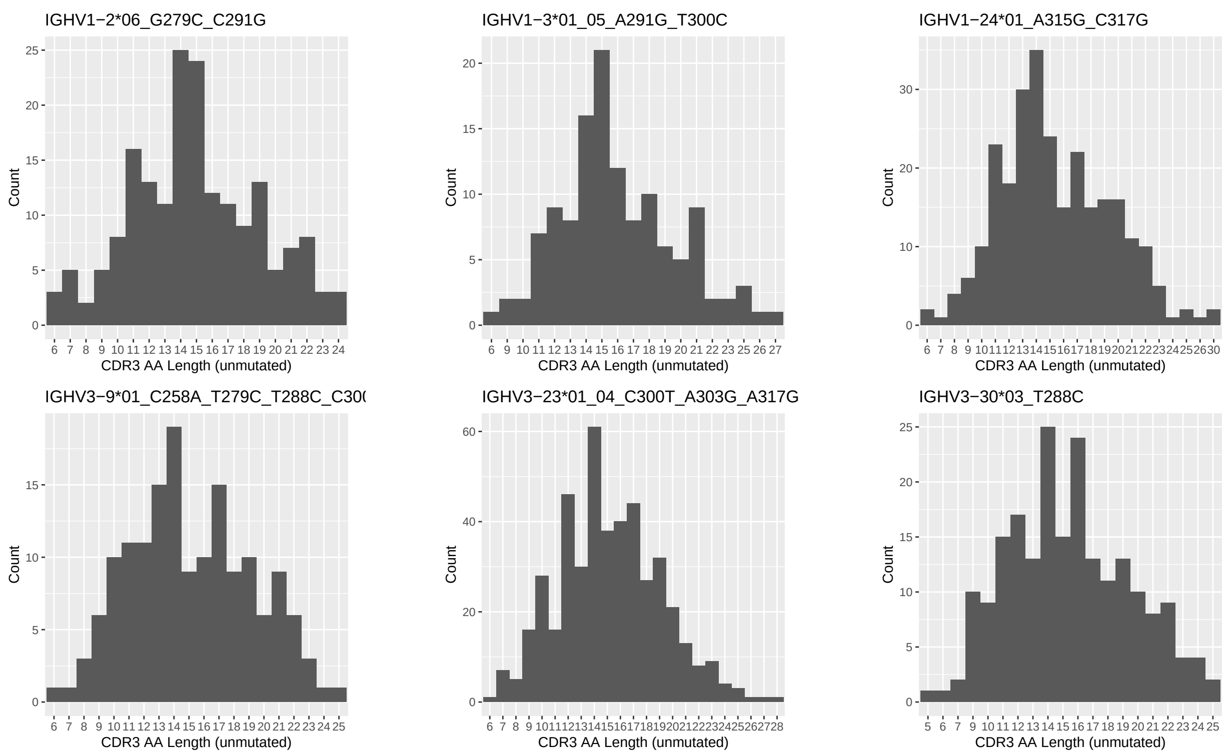


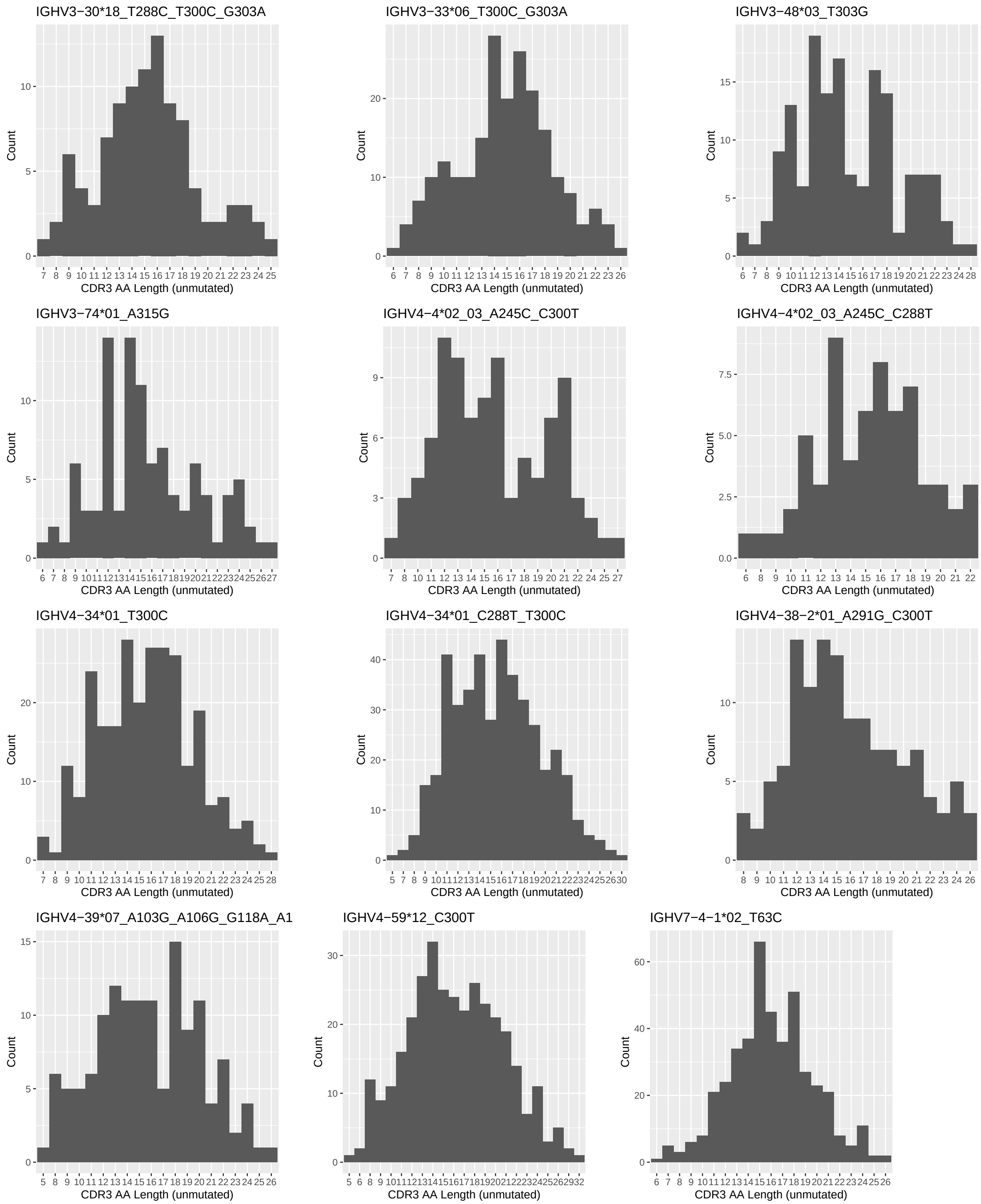
1.4 Final three nucleotides: frequency of each observed triplet



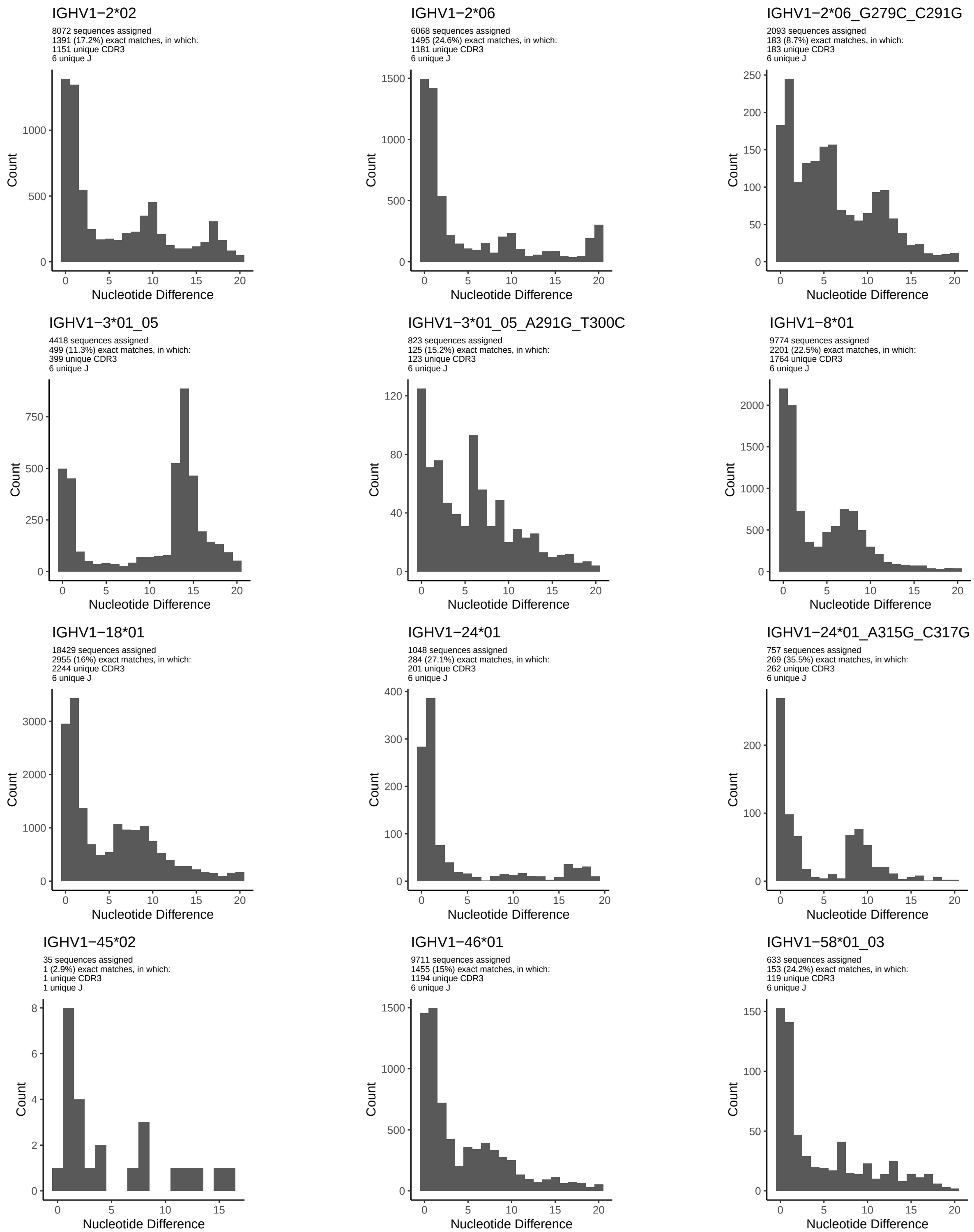


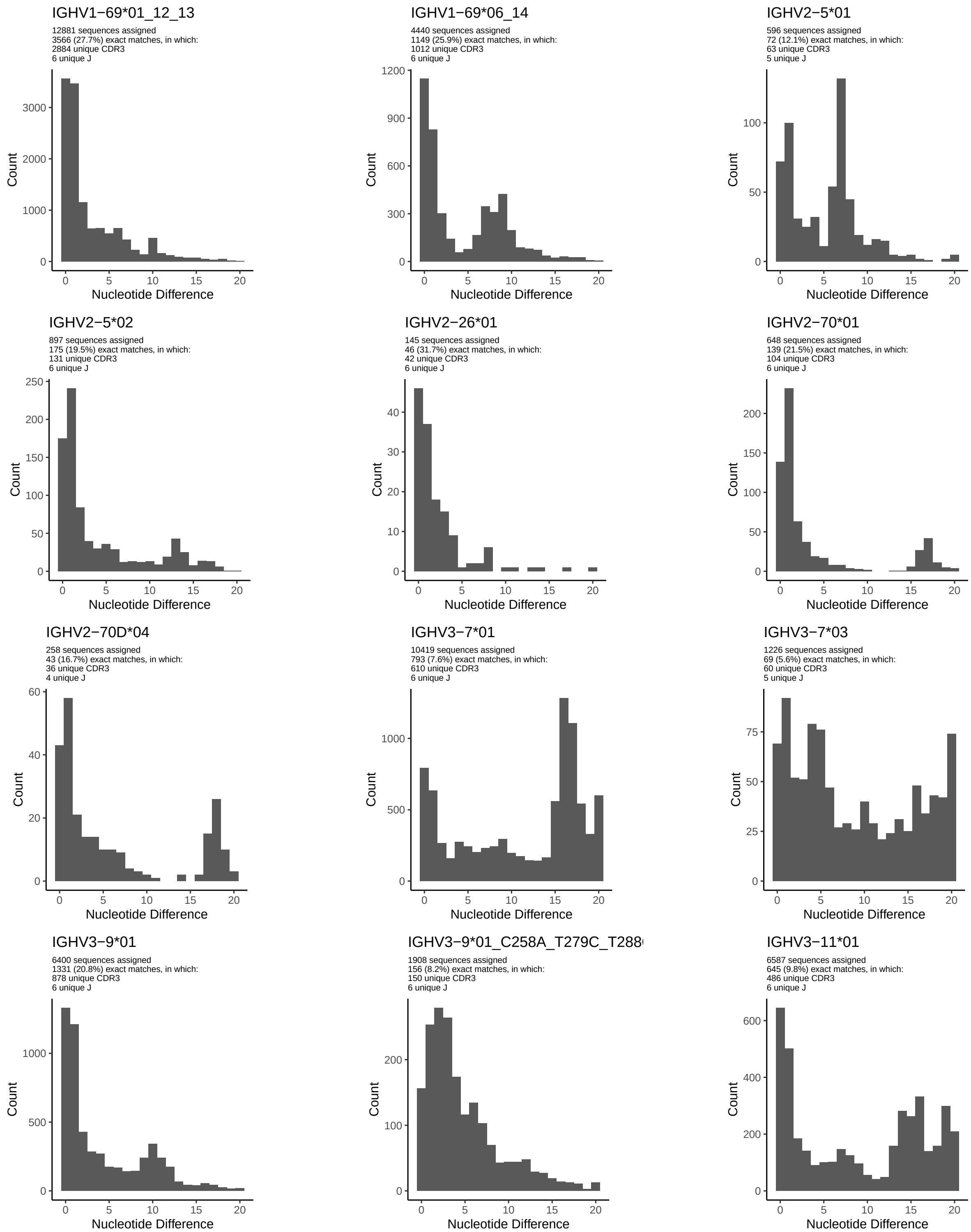
1.5 CDR3 length distribution, in assignments to novel alleles

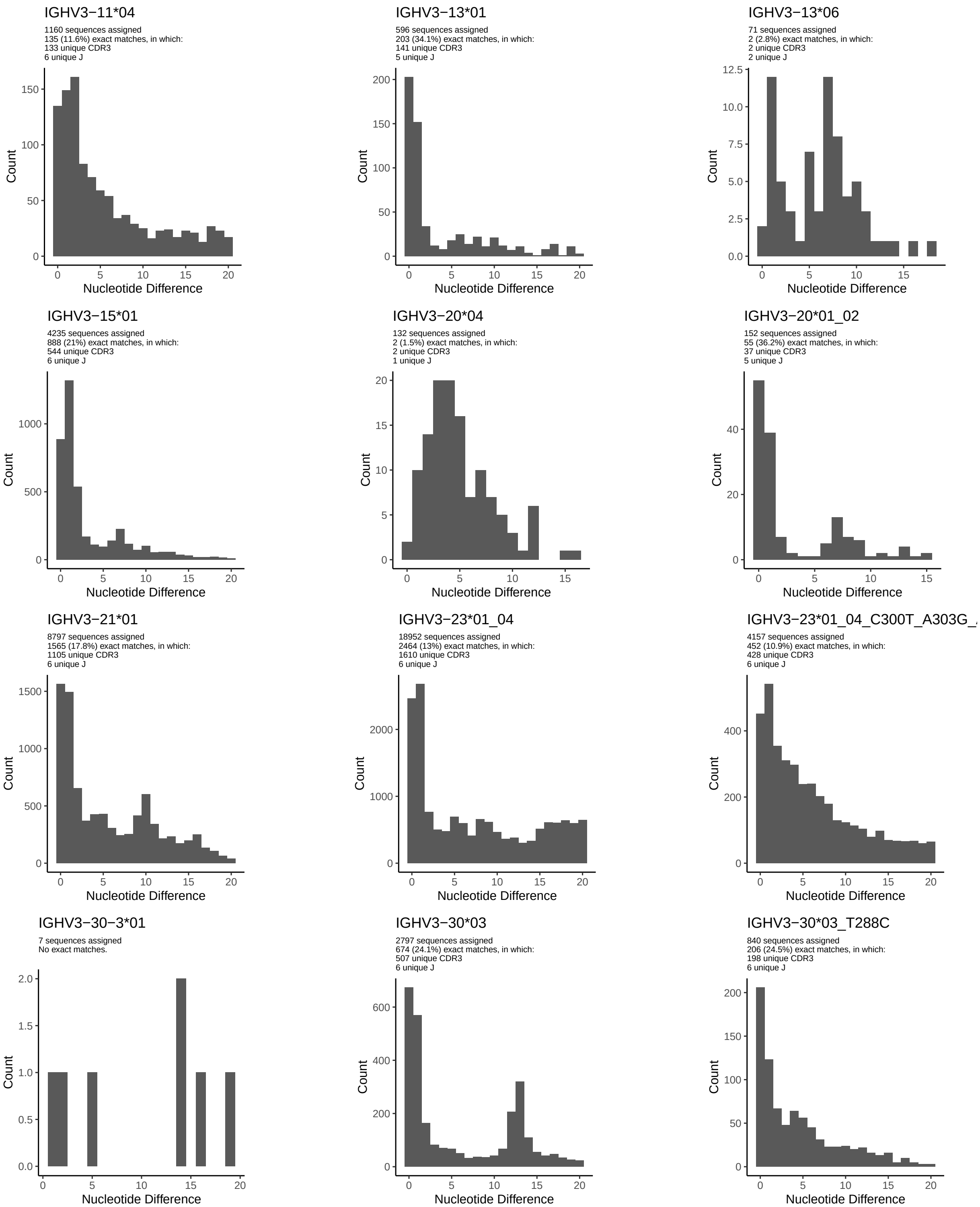


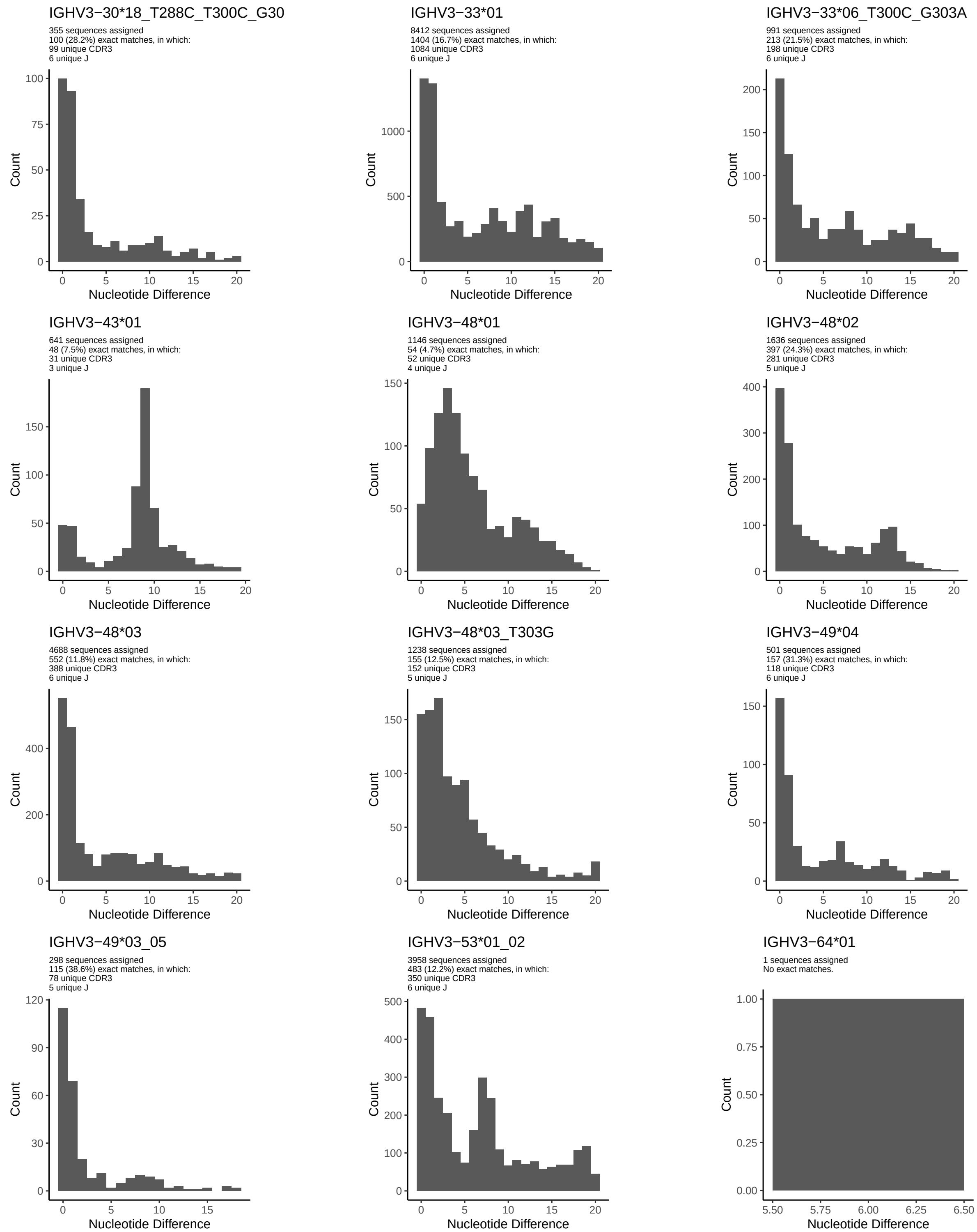

IGHV3−30*18_T288C_T300C_G303A
IGHV3−33*06_T300C_G303A
IGHV3−48*03_T303G
IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−4*02_03_A245C_C288T
IGHV4−34*01_T300C
IGHV4−34*01_C288T_T300C
IGHV4−38−2*01_A291G_C300T
IGHV4−39*07_A103G_A106G_G118A_A1
IGHV4−59*12_C300T
IGHV7−4−1*02_T63C
Count
CDR3 AA Length (unmutated)

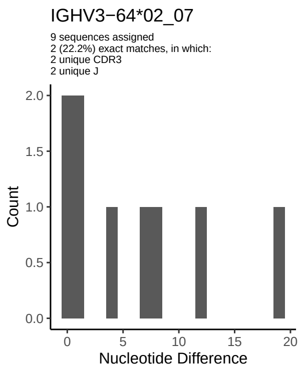
2 Variation from germline, in assignments to each allele

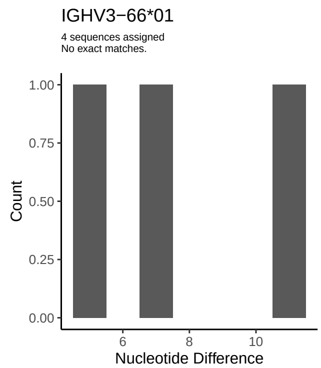


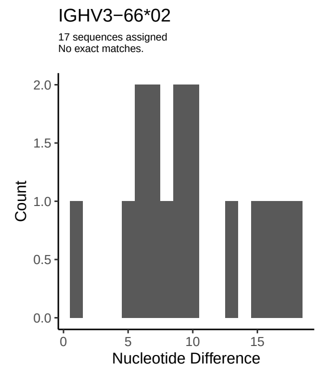

IGHV1−69*01_12_13
12881 sequences assigned
3566 (27.7%) exact matches, in which:
2884 unique CDR3
6 unique J
IGHV1−69*06_14
4440 sequences assigned
1149 (25.9%) exact matches, in which:
1012 unique CDR3
6 unique J
IGHV2−5*01
596 sequences assigned
72 (12.1%) exact matches, in which:
63 unique CDR3
5 unique J
IGHV2−5*02
897 sequences assigned
175 (19.5%) exact matches, in which:
131 unique CDR3
6 unique J
IGHV2−26*01
145 sequences assigned
46 (31.7%) exact matches, in which:
42 unique CDR3
6 unique J
IGHV2−70*01
648 sequences assigned
139 (21.5%) exact matches, in which:
104 unique CDR3
6 unique J
IGHV2−70D*04
258 sequences assigned
43 (16.7%) exact matches, in which:
36 unique CDR3
4 unique J
IGHV3−7*01
10419 sequences assigned
793 (7.6%) exact matches, in which:
610 unique CDR3
6 unique J
IGHV3−7*03
1226 sequences assigned
69 (5.6%) exact matches, in which:
60 unique CDR3
5 unique J
IGHV3−9*01
6400 sequences assigned
1331 (20.8%) exact matches, in which:
878 unique CDR3
6 unique J
IGHV3−9*01_C258A_T279C_T288
1908 sequences assigned
156 (8.2%) exact matches, in which:
150 unique CDR3
6 unique J
IGHV3−11*01
6587 sequences assigned
645 (9.8%) exact matches, in which:
486 unique CDR3
6 unique J
Count
Nucleotide Difference

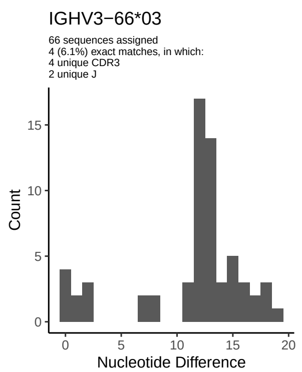

IGHV3−11*04
1160 sequences assigned
135 (11.6%) exact matches, in which:
133 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−13*01
596 sequences assigned
203 (34.1%) exact matches, in which:
141 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−13*06
71 sequences assigned
2 (2.8%) exact matches, in which:
2 unique CDR3
2 unique J
Count
Nucleotide Difference
IGHV3−15*01
4235 sequences assigned
888 (21%) exact matches, in which:
544 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−20*04
132 sequences assigned
2 (1.5%) exact matches, in which:
2 unique CDR3
1 unique J
Count
Nucleotide Difference
IGHV3−20*01_02
152 sequences assigned
55 (36.2%) exact matches, in which:
37 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−21*01
8797 sequences assigned
1565 (17.8%) exact matches, in which:
1105 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−23*01_04
18952 sequences assigned
2464 (13%) exact matches, in which:
1610 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−23*01_04_C300T_A303G_
4157 sequences assigned
452 (10.9%) exact matches, in which:
428 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−30−3*01
7 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV3−30*03
2797 sequences assigned
674 (24.1%) exact matches, in which:
507 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−30*03_T288C
840 sequences assigned
206 (24.5%) exact matches, in which:
198 unique CDR3
6 unique J
Count
Nucleotide Difference

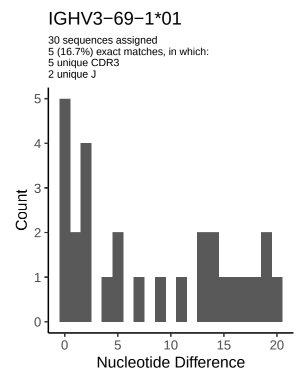

IGHV3−30*18_T288C_T300C_G30
355 sequences assigned
100 (28.2%) exact matches, in which:
99 unique CDR3
6 unique J
IGHV3−33*01
8412 sequences assigned
1404 (16.7%) exact matches, in which:
1084 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
991 sequences assigned
213 (21.5%) exact matches, in which:
198 unique CDR3
6 unique J
IGHV3−43*01
641 sequences assigned
48 (7.5%) exact matches, in which:
31 unique CDR3
3 unique J
IGHV3−48*01
1146 sequences assigned
54 (4.7%) exact matches, in which:
52 unique CDR3
4 unique J
IGHV3−48*02
1636 sequences assigned
397 (24.3%) exact matches, in which:
281 unique CDR3
5 unique J
IGHV3−48*03
4688 sequences assigned
552 (11.8%) exact matches, in which:
388 unique CDR3
6 unique J
IGHV3−48*03_T303G
1238 sequences assigned
155 (12.5%) exact matches, in which:
152 unique CDR3
5 unique J
IGHV3−49*04
501 sequences assigned
157 (31.3%) exact matches, in which:
118 unique CDR3
6 unique J
IGHV3−49*03_05
298 sequences assigned
115 (38.6%) exact matches, in which:
78 unique CDR3
5 unique J
IGHV3−53*01_02
3958 sequences assigned
483 (12.2%) exact matches, in which:
350 unique CDR3
6 unique J
IGHV3−64*01
1 sequences assigned
No exact matches.
Count
Nucleotide Difference

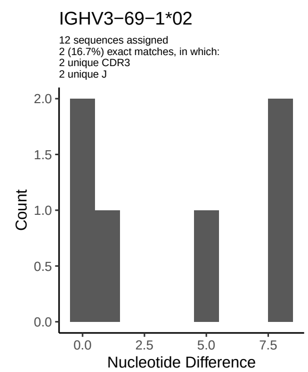

IGHV3−64*02_07
9 sequences assigned
2 (22.2%) exact matches, in which:
2 unique CDR3
2 unique J
Count
Nucleotide Difference

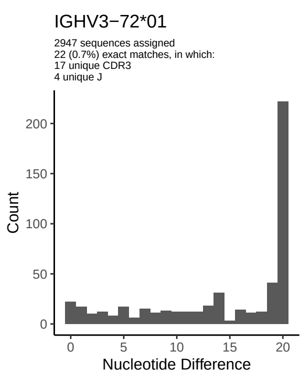

IGHV3−66*01
4 sequences assigned
No exact matches.
Count
Nucleotide Difference

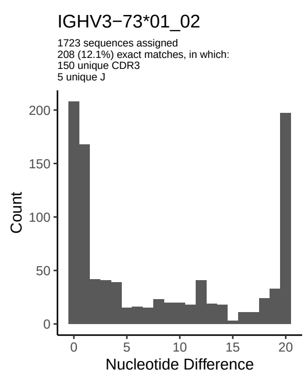

IGHV3−66*02
17 sequences assigned
No exact matches.
Count
Nucleotide Difference

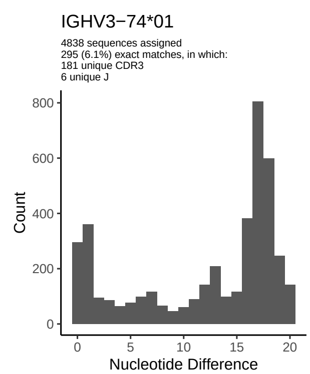

IGHV3−66*03
66 sequences assigned
4 (6.1%) exact matches, in which:
4 unique CDR3
2 unique J
Count
Nucleotide Difference

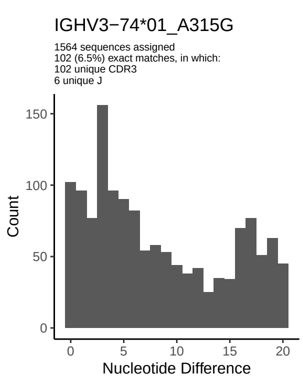

IGHV3−69−1*01
30 sequences assigned
5 (16.7%) exact matches, in which:
5 unique CDR3
2 unique J
Count
Nucleotide Difference

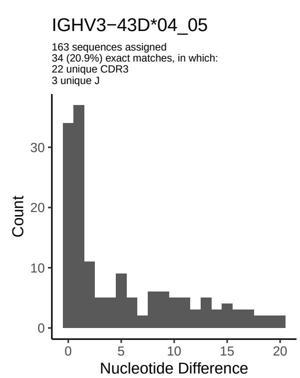

IGHV3−69−1*02
12 sequences assigned
2 (16.7%) exact matches, in which:
2 unique CDR3
2 unique J
Count
Nucleotide Difference

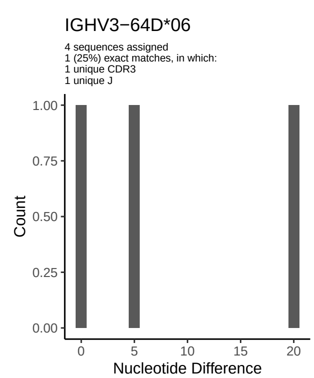

IGHV3−72*01
2947 sequences assigned
22 (0.7%) exact matches, in which:
17 unique CDR3
4 unique J
Count
Nucleotide Difference

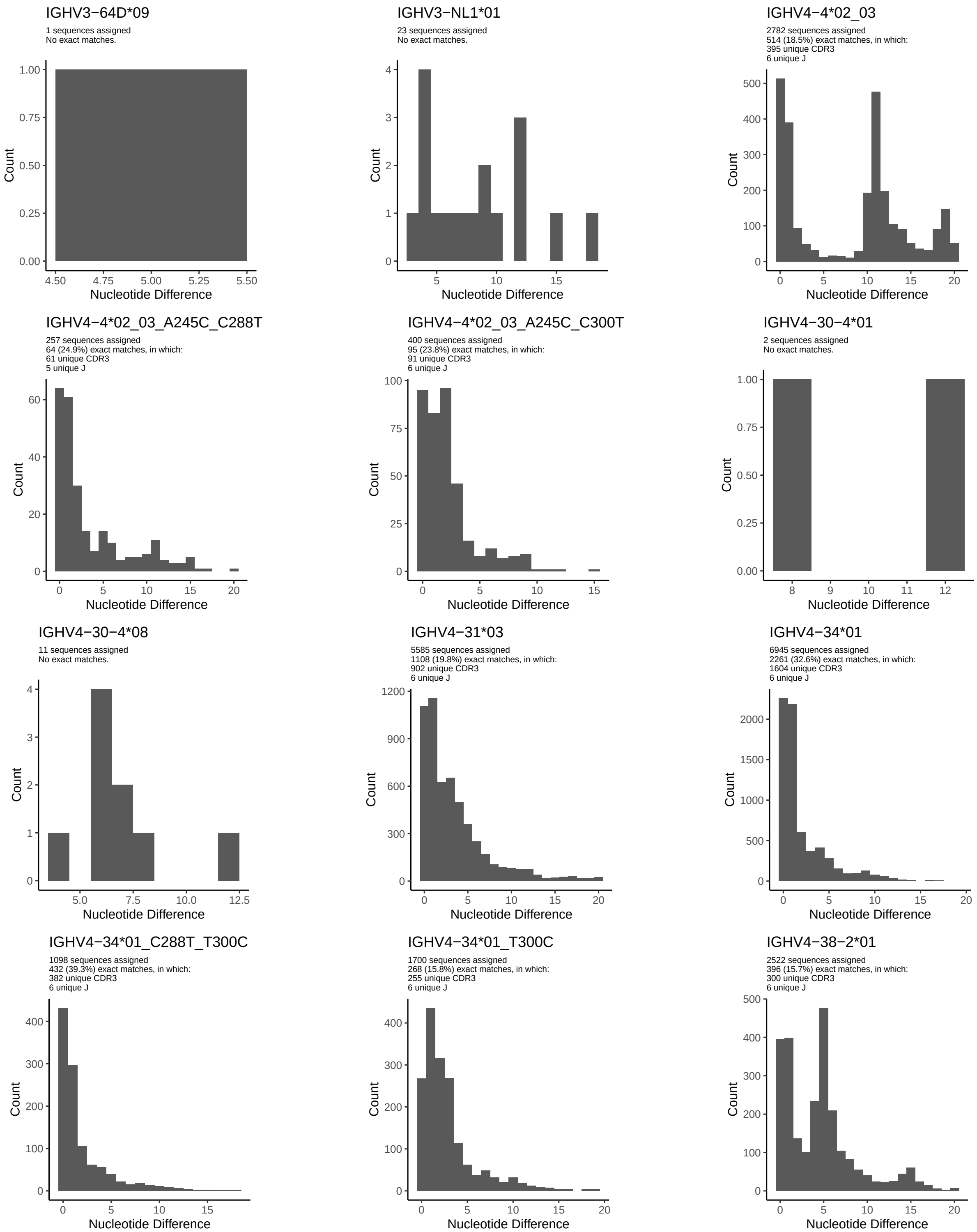

IGHV3−73*01_02
1723 sequences assigned
208 (12.1%) exact matches, in which:
150 unique CDR3
5 unique J
Count
Nucleotide Difference

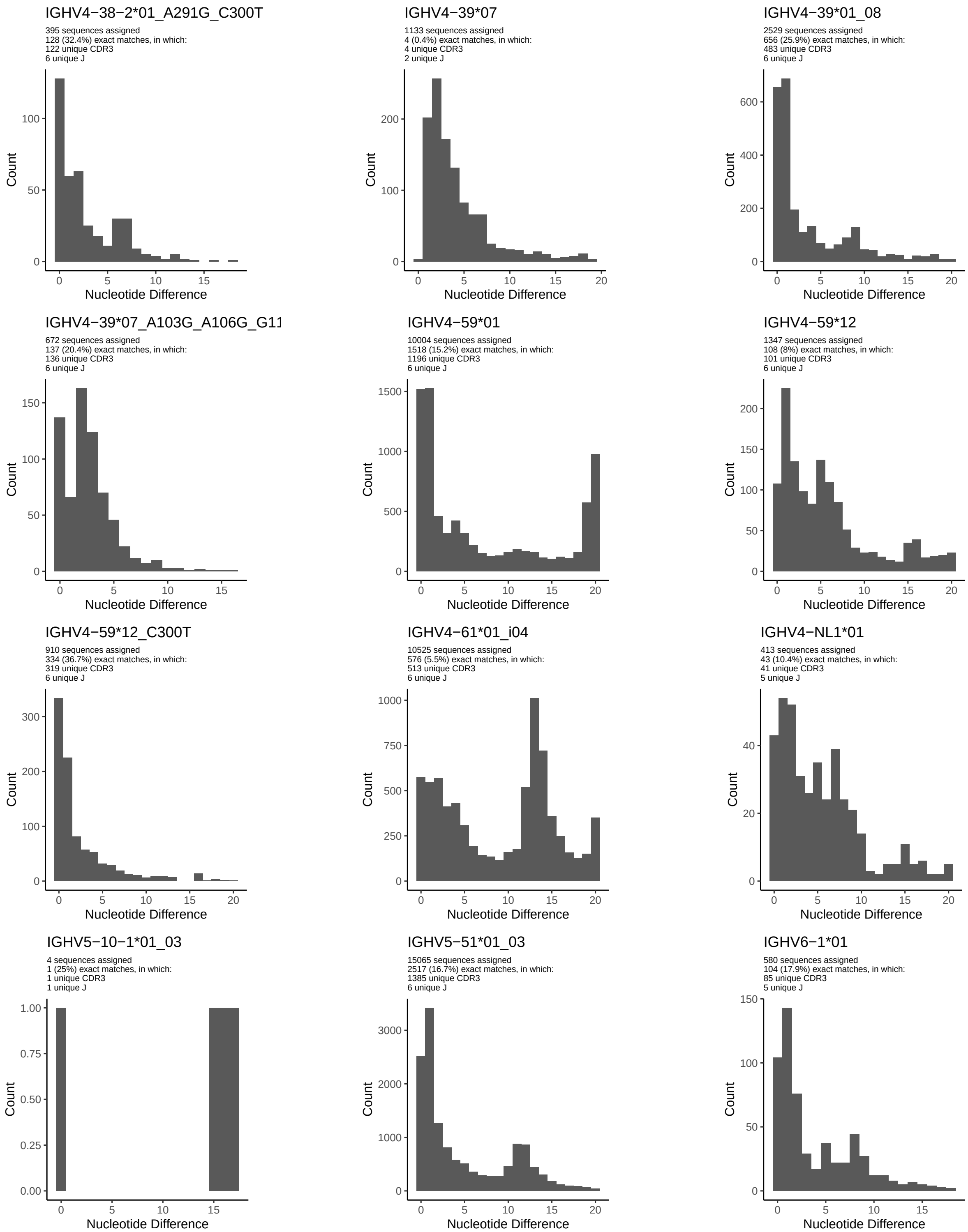

IGHV3−74*01
4838 sequences assigned
295 (6.1%) exact matches, in which:
181 unique CDR3
6 unique J
Count
Nucleotide Difference

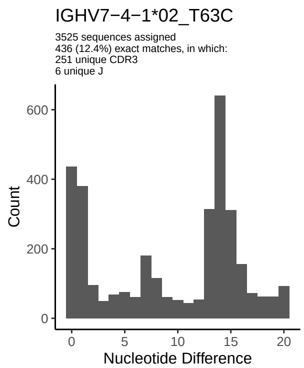

IGHV3−74*01_A315G
1564 sequences assigned
102 (6.5%) exact matches, in which:
102 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−43D*04_05
163 sequences assigned
34 (20.9%) exact matches, in which:
22 unique CDR3
3 unique J
Count
Nucleotide Difference

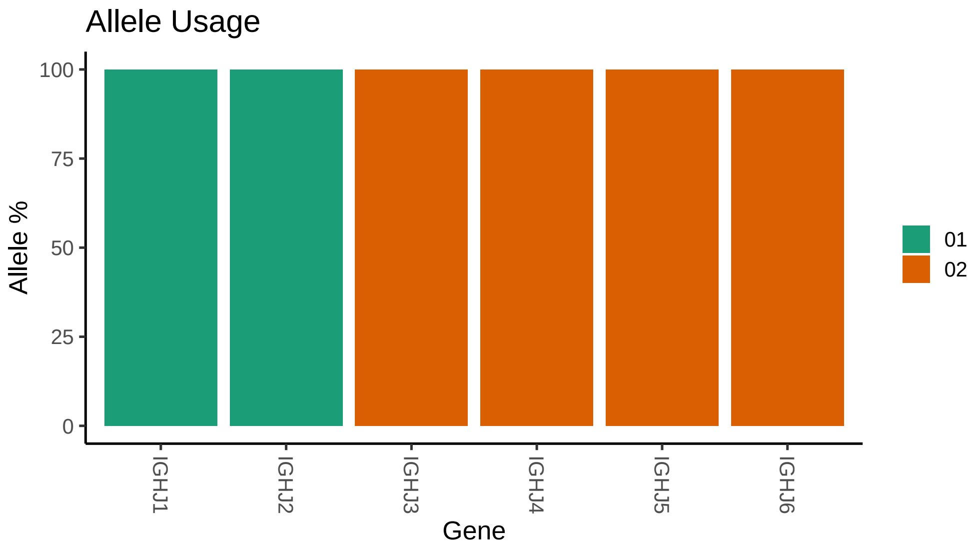

IGHV3−64D*06
4 sequences assigned
1 (25%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference


IGHV3−64D*09
1 sequences assigned
No exact matches.
IGHV3−NL1*01
23 sequences assigned
No exact matches.
IGHV4−4*02_03
2782 sequences assigned
514 (18.5%) exact matches, in which:
395 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C288T
257 sequences assigned
64 (24.9%) exact matches, in which:
61 unique CDR3
5 unique J
IGHV4−4*02_03_A245C_C300T
400 sequences assigned
95 (23.8%) exact matches, in which:
91 unique CDR3
6 unique J
IGHV4−30−4*01
2 sequences assigned
No exact matches.
IGHV4−30−4*08
11 sequences assigned
No exact matches.
IGHV4−31*03
5585 sequences assigned
1108 (19.8%) exact matches, in which:
902 unique CDR3
6 unique J
IGHV4−34*01
6945 sequences assigned
2261 (32.6%) exact matches, in which:
1604 unique CDR3
6 unique J
IGHV4−34*01_C288T_T300C
1098 sequences assigned
432 (39.3%) exact matches, in which:
382 unique CDR3
6 unique J
IGHV4−34*01_T300C
1700 sequences assigned
268 (15.8%) exact matches, in which:
255 unique CDR3
6 unique J
IGHV4−38−2*01
2522 sequences assigned
396 (15.7%) exact matches, in which:
300 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−38−2*01_A291G_C300T
395 sequences assigned
128 (32.4%) exact matches, in which:
122 unique CDR3
6 unique J
IGHV4−39*07
1133 sequences assigned
4 (0.4%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV4−39*01_08
2529 sequences assigned
656 (25.9%) exact matches, in which:
483 unique CDR3
6 unique J
IGHV4−39*07_A103G_A106G_G11
672 sequences assigned
137 (20.4%) exact matches, in which:
136 unique CDR3
6 unique J
IGHV4−59*01
10004 sequences assigned
1518 (15.2%) exact matches, in which:
1196 unique CDR3
6 unique J
IGHV4−59*12
1347 sequences assigned
108 (8%) exact matches, in which:
101 unique CDR3
6 unique J
IGHV4−59*12_C300T
910 sequences assigned
334 (36.7%) exact matches, in which:
319 unique CDR3
6 unique J
IGHV4−61*01_i04
10525 sequences assigned
576 (5.5%) exact matches, in which:
513 unique CDR3
6 unique J
IGHV4−NL1*01
413 sequences assigned
43 (10.4%) exact matches, in which:
41 unique CDR3
5 unique J
IGHV5−10−1*01_03
4 sequences assigned
1 (25%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV5−51*01_03
15065 sequences assigned
2517 (16.7%) exact matches, in which:
1385 unique CDR3
6 unique J
IGHV6−1*01
580 sequences assigned
104 (17.9%) exact matches, in which:
85 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV7−4−1*02_T63C
3525 sequences assigned
436 (12.4%) exact matches, in which:
251 unique CDR3
6 unique J
Count
600
400
200
0
0
5
10
15
20
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 033/M64 033/M64 033/M64 033/79/035f5ba2cb7738c80f36c3c
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 033/M64 033/M64 033/M64 033/79/035f5ba2cb7738c
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 033/M64 033/M64 033/M64 033/79/035f5ba2cb7738c80f36c
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 2 06: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 2 06_G279C_C291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 9 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_C258A_T279C_T288C_C300T_T301G_A315G_A317G: replacing with g
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 18_T288C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_A291G_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C_G189A_T19
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
```